



UNIVERSITY OF
LEICESTER

SCHOOL OF CMS

Coursework Report

Early Hospital Readmission Prediction using the 130 US Hospitals Diabetes Dataset

Group 46

Contents

1. Introduction	[2]
2. Dataset and Preprocessing	
2.1 Dataset overview	[2]
2.2 Data quality issues identified in the raw audit	[2]
2.3 Cleaning decisions and justification	[2]
2.4 Target variable and class imbalance	[3]
3. Exploratory Data Analysis	
3.1 Age and medication burden	[3]
3.2 Prior utilisation and discharge context	[4]
3.3 Numerical relationships and EDA summary	[4]
4. Initial Modelling	
4.1 Feature selection and preprocessing	[5]
4.2 Baseline models on imbalanced data	[5]
4.3 Stratified cross-validation on imbalanced data	[5]
4.4 Random undersampling	[5]
4.5 Balanced and tuned model comparison	[5]
4.6 Model interpretation and final initial-model selection	[6]
5. Distributed Modelling in PySpark	
5.1 Motivation for the distributed stage	[7]
5.2 Distributed preprocessing and feature engineering	[7]
5.3 Global distributed model	[7]
5.4 Comparison with the earlier-stage winner	[7]
5.5 Reflection on distributed processing	[7]
6. K-Means Clustering for Patient Subgroup Discovery	
6.1 Feature selection and preprocessing for clustering	[8]
6.2 Selection of the number of clusters	[8]
6.3 Cluster structure and visualisation	[8]
6.4 Cluster profile summary	[9]
6.5 Readmission rates across clusters	[9]
6.6 Interpretation of cluster meaning	[9]
7. Cluster-Based Local Classifiers	
7.1 Local model setup	[9]
7.2 Balanced local model results	[9]
7.3 Comparison of local and global models	[10]
7.4 When cluster-based classification is beneficial	[10]
8. Conclusion	[10]
References	[10]

1. Introduction

Hospital readmission within 30 days is an important healthcare outcome because it is associated with patient recovery, continuity of care, and hospital resource use. Predicting early readmission is especially relevant for patients with diabetes because they often have complex treatment needs, frequent contact with healthcare services, and multiple comorbidities. This project investigates whether records from the Diabetes 130 US hospitals dataset can be used to predict whether a hospital encounter is followed by readmission within 30 days of discharge.

The dataset includes demographic information, admission and discharge details, prior healthcare utilisation, laboratory related fields, diagnosis codes, and medication variables. The analysis began with a raw data audit and a systematic cleaning and preprocessing stage. This was followed by exploratory data analysis to identify patterns associated with early readmission. After that, three supervised classification models were developed and evaluated. These models were Logistic Regression, Random Forest, and Decision Tree. Class imbalance was addressed using random undersampling during the earlier modelling stage. The workflow was then extended through distributed modelling in PySpark, where grouped diagnosis categories were reintroduced. Finally, K Means clustering was used to explore possible patient subgroups, and local cluster based classifiers were compared with the global distributed model. The overall aim was to identify an effective predictive approach for early readmission and to examine whether distributed processing and cluster based modelling offered additional predictive or exploratory value.

2. Dataset and preprocessing

2.1 Dataset overview

The project used the diabetes readmission dataset, which contains ten years of clinical care records collected from 130 US hospitals and integrated delivery networks. The raw dataset included 101,766 records at the encounter level and 50 variables. Each row represented a hospital encounter rather than a unique patient, so repeated patients were possible. The original target variable, readmitted, contained three categories: <30, >30, and NO. It was later converted into a binary outcome for early readmission prediction. Before any cleaning was carried out, the raw data were audited to examine the dataset structure, missing and irregular values, repeated encounters, and the distribution of the target variable. The data were initially loaded as text because several hospital fields, including admission_type_id, discharge_disposition_id, and admission_source_id, were coded categories rather than true numeric variables. The diagnosis fields diag_1, diag_2, and diag_3 also required care because they contained mixed ICD style values. The IDS mapping file was therefore used to interpret these coded hospital variables correctly, particularly when identifying discharge codes associated with expired outcomes.

2.2 Data quality issues identified in the raw audit

The raw data audit identified several important quality issues before preprocessing began. Missing values were represented inconsistently, with the placeholder ? appearing mainly in columns such as weight, medical_specialty, and payer_code, while smaller amounts of missingness were also present in race and the diagnosis fields. The laboratory related variables max_glu_serum and A1Cresult were dominated by the category None, which required careful interpretation rather than automatic treatment as standard missing data. The target variable readmitted contained the three raw classes <30, >30, and NO, and early readmission within 30 days formed the minority class, showing that the dataset was imbalanced. The audit also showed that encounter_id and patient_nbr behaved as identifier variables rather than general predictive features, and that the data were encounter based rather than patient based because many patients appeared more than once even though no exact duplicate full rows were found. In addition, examide and citoglipton showed no variation, several medication variables had near zero variance with overwhelmingly dominant No values, and the diagnosis fields diag_1, diag_2, and diag_3 contained mixed ICD style codes that could not be treated as ordinary numeric variables.

2.3 Cleaning decisions and justification

The aggressive cleaning stage began by standardising the placeholder ? to missing values so that missingness could be handled consistently. The identifier columns encounter_id and patient_nbr were removed, as were the constant variables examide and citoglipton. The high missingness columns weight, medical_specialty, and payer_code were excluded from the aggressive cleaned dataset. The target variable readmitted was converted into a binary outcome called readmitted_binary, where <30 was treated as the positive class and both >30 and NO were grouped as the negative class. Count based utilisation variables were converted into numeric form where needed, expired discharge outcomes identified through the IDS mapping file were removed, and eight medication variables with near zero variance were also excluded. The diagnosis fields diag_1, diag_2, and diag_3 were retained in their ICD style form, while feature scaling was deferred to the later modelling pipelines. After these steps, the aggressive cleaned dataset contained 100,114 rows and 36 columns.

These decisions were intended to improve data quality without discarding useful clinical information. Standardising ? values was necessary because non-standard missing markers would otherwise prevent consistent auditing and preprocessing. The identifier columns were removed because they label encounters or patients rather than describing generalisable clinical risk, while the constant columns had no predictive value. The three high-missing columns were excluded because simple imputation would have been dominated by missing data rather than observed information. Expired discharge outcomes were removed because a deceased patient cannot later be readmitted, so keeping these rows would have made the target logically inconsistent. The near zero variance medication columns were excluded because they offered very little learnable variation. At the same time, the diagnosis fields were not forced into numeric form because their mixed ICD style values would have made naïve conversion misleading, and scaling was postponed to leakage safe modelling pipelines so that transformations would be fitted only on the training data. Outliers were reviewed but not removed at this stage because extreme utilisation values were clinically plausible rather than obvious errors and removing them early could have discarded high risk heavy utilisation patients; instead, they were handled later in the distributed modelling.

Item	Value
Raw dataset shape	101,766 rows x 50 columns
Aggressively cleaned dataset shape	100,114 rows x 36 columns
Target conversion	readmitted_binary: <30 = 1, >30 and NO = 0
High-missingness columns removed	weight, medical_specialty, payer_code
Other removals	8 near-zero-variation medication columns; expired discharge outcomes

Table 1. Summary of the main cleaning outcomes

2.4 Target variable and class imbalance

The original target variable, readmitted, was converted into a binary outcome to match the coursework objective of predicting early readmission within 30 days. In this formulation, <30 was treated as the positive class, while >30 and NO were grouped together as the negative class because both represent outcomes that do not correspond to readmission within the 30 day window. After cleaning and target conversion, the dataset was strongly imbalanced, with the negative class accounting for 88.66% of cases and the positive class accounting for 11.34%. This imbalance is shown clearly in Figure 1. It also meant that accuracy alone would be a misleading evaluation measure, since a model could achieve high accuracy simply by predicting the majority class. The class distribution therefore made it necessary to use more informative metrics such as precision, recall, F1 score, and ROC AUC, and justified the use of random undersampling in the initial modelling stage.

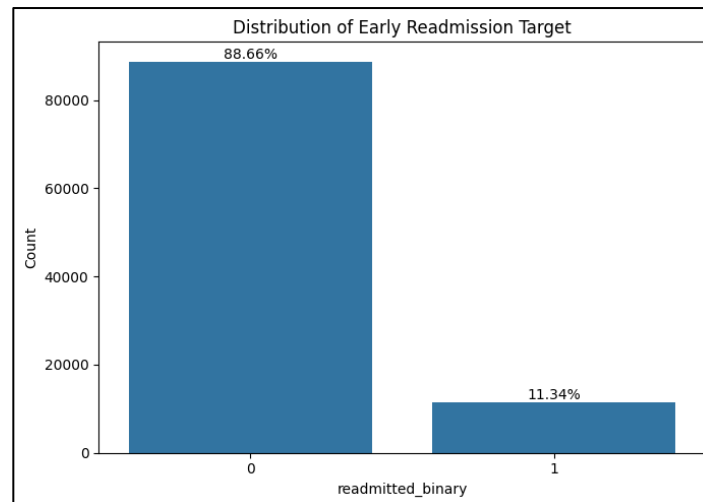


Figure 1. Distribution of the binary early-readmission target.

3. Exploratory data analysis

Exploratory analysis was performed on the aggressively cleaned dataset to understand which variables appeared most relevant to early readmission before formal modelling. The analysis focused on class balance, utilisation history, age patterns, medication burden, and administrative context such as admission and discharge pathways. The central conclusion from this stage was that no single variable cleanly separated the classes, and that readmission risk was driven more by combinations of prior utilisation, treatment burden, and discharge context than by any one dominant feature.

3.1 Age and medication burden

Readmission varied across age groups, but the pattern was not strictly monotonic. The highest observed rate was 14.30% in the [20–30) age band. This elevated rate may partly reflect the smaller sample size in that age band. Most adult and older groups clustered around roughly 10–12%. This suggests that age contains some predictive signal, but not in a simple form where risk increases steadily with age. Medication burden showed a somewhat clearer relationship with early readmission. The readmission rate increased across the lower and middle medication groups, reaching 13.17% in the 21–30 group and 13.06% in the 31–50 group, before declining slightly in the highest medication category. This indicates that treatment complexity is associated with readmission risk, although the relationship is not perfectly linear. These findings supported the inclusion of age and medication related variables in the later modelling stage.

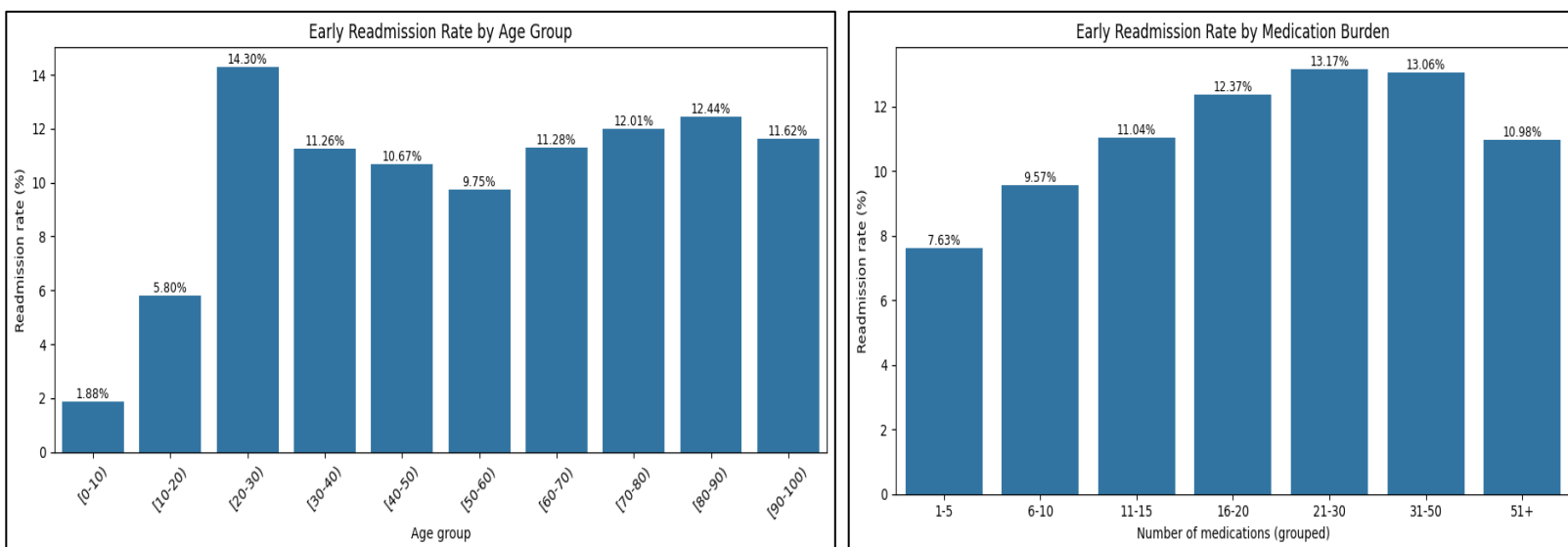


Figure 2. Early readmission rate by age group (left) and grouped medication burden (right).

3.2 Prior utilisation and discharge context

Prior inpatient utilisation provided the strongest single signal in the exploratory analysis. Patients with zero prior inpatient visits had an early readmission rate of 8.56%, whereas patients with more frequent previous inpatient admissions showed much higher rates, rising above 30% and reaching over 40% for the highest counts displayed in the chart. This pattern is clinically plausible because repeated inpatient use is often a marker of unresolved or recurrent health problems and therefore suggests a higher likelihood of short term readmission. Discharge destination also carried meaningful signal. Patients discharged to home had a readmission rate of 9.30%, compared with 16.07% for discharge to another short term hospital, 20.86% for discharge to an inpatient care institution, 27.70% for discharge to rehabilitation, and 36.69% for discharge to a psychiatric hospital among categories with at least 100 observations.

These findings indicate that both prior healthcare utilisation and post discharge pathway are important markers of patient complexity. They also helped justify the retention of utilisation and discharge related variables as key predictors in the modelling stage, particularly number_inpatient and discharge_disposition_id. More broadly, the strength of these patterns supported the later decision to test whether subgrouping patients through clustering might be useful, since the exploratory analysis suggested that readmission risk varied across clinically different utilisation profiles.

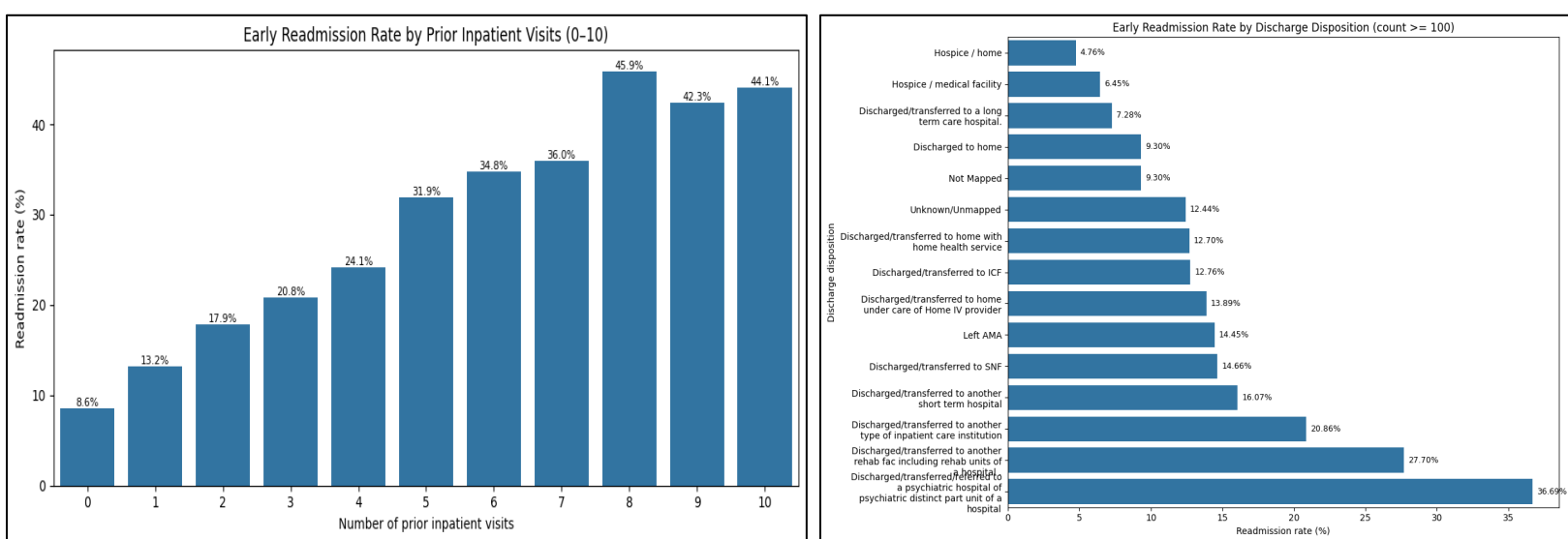


Figure 3. Early readmission rate by prior inpatient utilisation (left) and discharge disposition (right).

3.3 Correlation structure and EDA summary

A correlation review of the key numeric variables showed that the strongest linear relationships were between time_in_hospital and num_medications (0.46), num_procedures and num_medications (0.38), and time_in_hospital and num_lab_procedures (0.32). By contrast, the numeric correlations with the target were all relatively weak, with number_inpatient showing the strongest positive relationship at around 0.17. This suggested that early readmission was not dominated by any single numeric predictor. Instead, the EDA indicated that readmission risk was shaped more by combinations of prior utilisation, treatment burden, and discharge context than by one dominant feature.

Taken together, the exploratory analysis showed that the strongest signals in the dataset came from prior inpatient history, discharge pathway, and measures of treatment complexity, while age and the key numeric variables provided more moderate supporting information. These findings suggested that simple one-variable reasoning would be insufficient for the prediction task. This directly supported the use of multivariate modelling in the next stage and justified comparing Logistic Regression with tree-based methods such as Decision Tree and Random Forest, which are better able to capture non-linear interactions between variables.

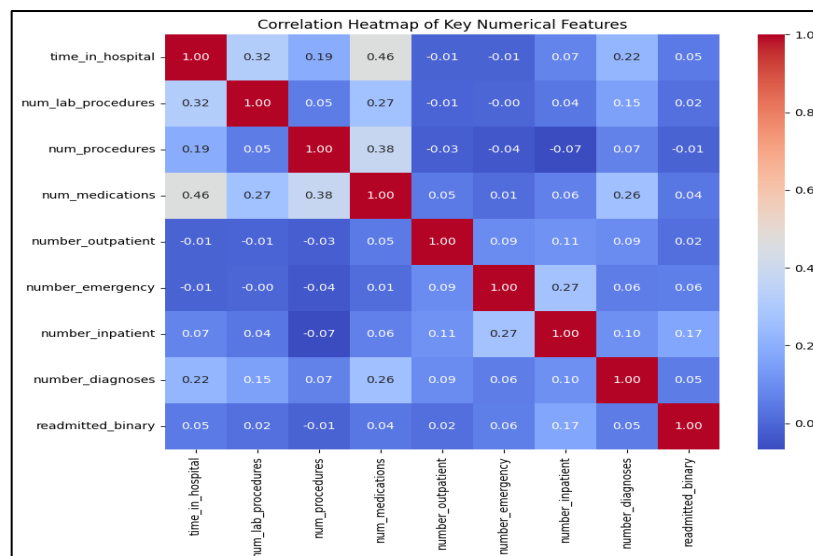


Figure 4. Correlation heatmap of the main numerical features.

4. Initial model building and evaluation

4.1 Feature selection and preprocessing

The initial modelling stage used the aggressively cleaned dataset reloaded from cleaned_aggressive.csv. The target variable was readmitted_binary, and the original readmitted field was excluded to avoid leakage. The diagnosis fields diag_1, diag_2, and diag_3 were also excluded from the baseline stage because their mixed ICD style format and high cardinality made them unsuitable for direct use in the initial classical models. These fields were instead reintroduced later in the distributed stage through grouped diagnosis categories. After excluding these fields and any temporary helper variables, the baseline modelling stage used 31 predictors. The data were then split using an 80/20 stratified train test split, producing 80,091 training rows and 20,023 test rows while preserving the class distribution. The initial classifiers and preprocessing workflow were implemented using the scikit learn framework (Pedregosa et al., 2011).

4.2 Baseline models on imbalanced data:

An 80/20 stratified train test split was used to preserve the class distribution in both subsets, producing **80,091** training rows and **20,023** test rows. A dummy classifier was first evaluated to provide a trivial baseline under the original class imbalance. As expected, it achieved high accuracy simply by predicting the majority class, but it produced zero precision, recall, and F1 score for the minority class. This confirmed that accuracy alone was misleading for this problem. Three classifiers were then trained on the imbalanced training data: Logistic Regression, Random Forest, and Decision Tree. All three models achieved similarly high accuracy, but minority class detection remained very weak. Logistic Regression produced the strongest ROC AUC among the imbalanced baselines, while the Decision Tree gave slightly better recall and F1 score than the other two models. However, none of the imbalanced models performed well enough on early readmission detection to be considered satisfactory. This motivated the use of cross-validation and, later, explicit imbalance handling.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Decision Tree	0.8856	0.4074	0.0194	0.0370	0.6419
Logistic Regression	0.8864	0.4706	0.0141	0.0274	0.6444
Random Forest	0.8867	0.5227	0.0101	0.0199	0.6315
Dummy	0.8866	0.0000	0.0000	0.0000	NaN

Table 2. Performance of baseline models on the imbalanced dataset

4.3 Cross-validation strategy and justification:

Stratified k-fold cross-validation was then applied to the training data only in order to obtain a more reliable estimate of model performance before any imbalance handling was introduced. Stratification was important because it preserved the minority class proportion in each fold, which reduced the risk of unstable evaluation under the strong class imbalance in the dataset. Performing cross-validation on the training set only also avoided leakage from the held-out test set, so the final test results remained a fair measure of generalisation. The cross-validation results were consistent with the earlier train-test findings and confirmed that minority class detection remained weak across all three baseline models. Among the imbalanced classifiers, the Decision Tree produced the strongest recall and F1 score, while Logistic Regression and Decision Tree produced similar ROC AUC values. This showed that the poor minority class performance was not simply due to one particular split of the data, and it strengthened the case for explicit imbalance handling in the next stage.

4.4 Handling class imbalance with random undersampling:

Class imbalance was then addressed using random undersampling, in line with the coursework requirement and implemented through the imbalanced-learn toolkit (Lemaître et al., 2017). Before undersampling, the training data contained 71,005 majority class cases and 9,086 minority class cases. Random undersampling reduced the majority class to 9,086 cases so that the balanced training set contained 18,172 rows in total. The test set was left unchanged in order to preserve a realistic evaluation setting. This was important because balancing only the training data allowed the models to learn from an equal class distribution while preserving realistic test conditions. The purpose of this step was to reduce the tendency of the classifiers to favour the majority class and to improve detection of early readmission in the minority class. This stage therefore tested whether improving class balance in the training data would lead to more meaningful recall and F1 scores for early readmission detection.

Stage	Class 0	Class 1
Before undersampling	71,005	9,086
After undersampling	9,086	9,086

Table 3. Training class distribution before and after random undersampling

4.5 Balanced and tuned model comparison:

Random undersampling substantially improved minority class detection across all three model families. Logistic Regression with random undersampling achieved a recall of 0.5121 and an F1 score of 0.2586, while Random Forest achieved a higher recall of 0.5883 with an F1 score of 0.2532. Decision Tree also improved markedly, reaching a recall of 0.5694 and an F1 score of 0.2495. These results showed that balancing the training data had a much greater effect on minority class performance than simply changing the classifier under the original imbalance.

Hyperparameter tuning produced the clearest improvement for the Random Forest model. The tuned Random Forest achieved a recall of 0.6306, an F1 score of 0.2690, and a ROC AUC of 0.6590, making it the strongest model in the initial modelling stage. The tuned Logistic Regression and tuned Decision Tree produced only modest changes relative to their balanced versions, while the Random Forest benefited most from tuning. Overall, these results showed that the combination of random undersampling and model tuning produced a much stronger balance between minority class detection and overall discrimination than the earlier imbalanced models.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression + RUS	0.6669	0.1729	0.5121	0.2586	0.6425
Random Forest + RUS	0.6065	0.1613	0.5883	0.2532	0.6375
Decision Tree + RUS	0.6116	0.1598	0.5694	0.2495	0.6313
Tuned Logistic + RUS	0.6665	0.1728	0.5125	0.2585	0.6425
Tuned Random Forest + RUS	0.6114	0.1710	0.6306	0.2690	0.6590
Tuned Decision Tree + RUS	0.6567	0.1727	0.5346	0.2610	0.6368

Table 4. Performance of balanced and tuned models after random undersampling

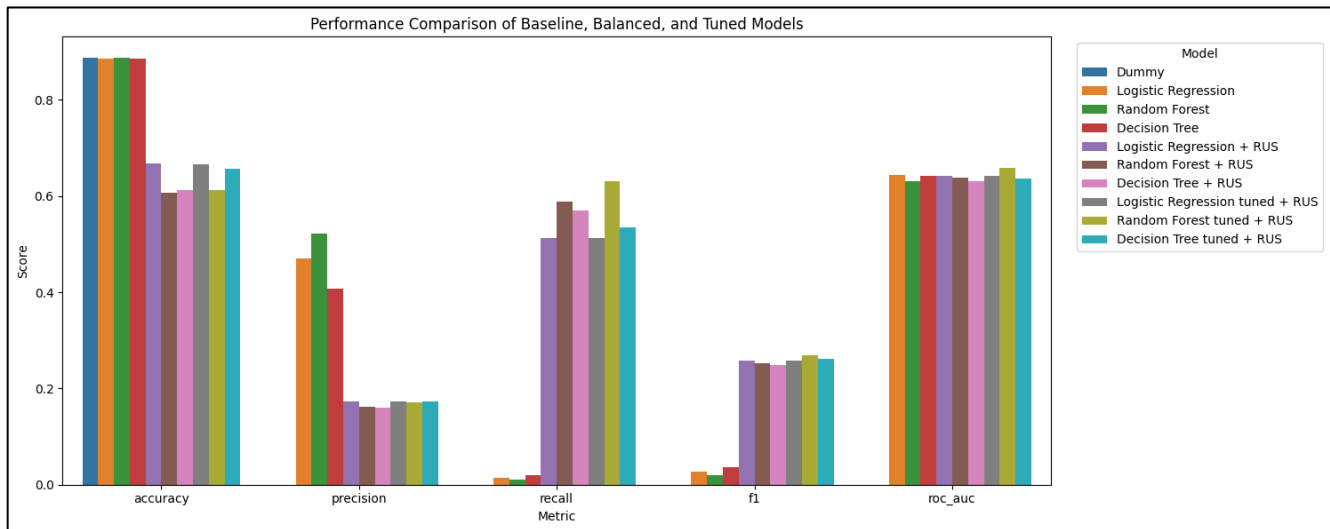


Figure 5. Visual comparison of baseline, balanced, and tuned model performance.

4.6 Model strengths, limitations, and interpretation:

Logistic Regression provided the clearest and most interpretable baseline, but its linear decision boundary limited its ability to capture more complex relationships between prior utilisation, treatment burden, and discharge context. It was still useful because it offered a transparent benchmark against which the tree based models could be compared. Decision Tree produced slightly stronger recall than the other imbalanced baselines and remained easy to interpret, but its performance was less stable and weaker than the ensemble alternative after balancing and tuning. Random Forest provided the strongest overall trade off because it was better able to capture non-linear interactions between predictors, which became especially important once class imbalance had been addressed. This helps explain why the Random Forest benefited most from tuning, while the tuned Logistic Regression and tuned Decision Tree changed only modestly relative to their balanced versions. After tuning, the Random Forest achieved the best overall performance, with recall 0.6306, F1 score 0.2690, and ROC AUC 0.6590, making it the final winner of the initial modelling stage.

The feature interpretation results were also consistent with the earlier exploratory analysis. In the Decision Tree, number_inpatient was the strongest predictor with an importance of 0.447, which matched the strong EDA signal from prior inpatient utilisation. In the Random Forest, the most important variables included num_lab_procedures (0.136), num_medications (0.116), and time_in_hospital (0.081), indicating that treatment burden and hospital use contributed strongly to prediction. These importance patterns support the broader conclusion that early readmission was influenced less by any single demographic feature and more by a combination of utilisation history, treatment complexity, and clinical burden. They also show that the strongest predictors in the final model were aligned with the most informative patterns identified earlier in the EDA stage. This strengthens the overall consistency of the modelling results and helps explain why the tuned Random Forest outperformed the simpler linear baseline.

5. Distributed analysis with PySpark

5.1 Motivation for the distributed stage:

The distributed stage was introduced to move beyond the limitations of the aggressively cleaned baseline dataset and to retain more information from the original raw data. In particular, the earlier modelling stage had excluded the diagnosis fields because their mixed ICD style format and high cardinality made them unsuitable for direct use in the initial classical models. The distributed workflow therefore returned to the full raw dataset so that diagnosis information could be reintroduced in a more structured way through grouped diagnosis categories, while missing values and outliers could be handled more selectively than in the earlier aggressive cleaning stage. This stage was intended not only to test whether fuller data retention could improve prediction, but also to demonstrate a distributed approach to preprocessing and model development using PySpark (Zaharia et al., 2016).

5.2 Distributed preprocessing and data preparation:

The distributed stage used the full raw dataset in Spark, which contained 101,766 rows and 50 columns. Missing and irregular values were reviewed again within the Spark workflow so that diagnosis information and other fields removed from the aggressive cleaned dataset could be reconsidered more selectively. The diagnosis variables were reintroduced through grouped categories, which reduced the complexity of the original ICD-style codes while still preserving clinically relevant information. This produced broader diagnosis groups such as circulatory, respiratory, digestive, diabetes, injury, genitourinary, musculoskeletal, neoplasms, other, and missing.

Outliers in the main utilisation and treatment variables were reviewed explicitly rather than removed through row deletion. This was important because extreme values in variables such as number_outpatient, number_emergency, number_inpatient, num_medications, and time_in_hospital were clinically plausible and could themselves indicate higher-risk patients. A separate modelling-oriented step explored capping and scaling for selected numerical variables, while the final Spark preprocessing pipeline applied Imputer, StringIndexer, OneHotEncoder, VectorAssembler, and StandardScaler to the full distributed dataset. This resulted in a final feature space of 147 variables for the global Spark model.

5.3 Global distributed model:

The initial global distributed model was built using Spark Logistic Regression on the full preprocessed feature set. The data were split into 81,565 training rows and 20,201 test rows. Although the unweighted model produced strong overall accuracy, its minority class performance was very weak, with recall of only 0.0147 and an F1 score of 0.0286. This showed that simply moving to a distributed model and retaining more raw information did not automatically solve the class imbalance problem. To address this, a weighted Spark Logistic Regression model was then introduced so that the minority class received greater importance during training while preserving the full dataset.

The weighted global model substantially improved minority class detection. It achieved precision 0.1695, recall 0.5250, F1 score 0.2563, and ROC AUC 0.6451, making it the strongest model within the distributed stage. The main improvement was in recall, which increased sharply compared with the unweighted version, while F1 score also improved considerably. This showed that class weighting was a much more effective strategy than leaving the distributed model unadjusted under the original imbalance.

5.4 Comparison with the earlier-stage winner:

The weighted Spark Logistic Regression was then compared with the strongest model from the initial modelling stage, namely the tuned Random Forest with random undersampling. This comparison showed that the weighted distributed model substantially improved on the unweighted Spark version, but it still remained slightly weaker than the earlier-stage winner overall. The weighted Spark model achieved recall 0.5250, F1 score 0.2563, and ROC AUC 0.6451, whereas the tuned Random Forest achieved recall 0.6306, F1 score 0.2690, and ROC AUC 0.6590. This means that the weighted distributed model became the best approach within the Spark stage, but the earlier tuned Random Forest still provided the strongest overall predictive performance. The distributed stage therefore added value mainly through fuller data retention, richer preprocessing, and scalable modelling, rather than by producing the single best final classifier.

Model	Precision	Recall	F1	ROC-AUC
Unweighted Spark Logistic Regression	0.4783	0.0147	0.0286	0.6437
Weighted Spark Logistic Regression	0.1695	0.5250	0.2563	0.6451
Earlier-stage winner: Tuned Random Forest + RUS	0.1710	0.6306	0.2690	0.6590

Table 5. Comparison of global distributed models and the earlier-stage benchmark

5.5 Reflection on distributed processing:

The distributed stage showed that moving to PySpark did not automatically produce the best predictive performance, but it did provide important practical advantages. PySpark made it possible to work with the full raw dataset, retain more information than in the earlier aggressive cleaning stage, and build a scalable preprocessing pipeline around imputation, encoding, vector assembly, and standard scaling. Its execution model also differs from a standard single machine workflow because transformations are evaluated lazily, organised through a directed acyclic graph, and applied across data partitions rather than being processed entirely in memory on one machine. This makes the pipeline more suitable for larger scale datasets, even when the gain on a dataset of this size is more methodological than computational.

In this project, the main value of distributed processing was therefore not a decisive improvement in final predictive performance, but the ability to preserve richer information, reintroduce diagnosis categories in a structured way, and demonstrate a workflow that would scale more naturally to larger healthcare data. The weighted Spark model became the strongest distributed model, but the earlier tuned Random Forest still remained slightly stronger overall. This suggests that distributed processing should be viewed here as an important extension of the analytical workflow rather than as a guaranteed route to the single best classifier.

6. Clustering analysis

6.1 Feature selection and preprocessing for clustering:

Clustering was performed on a reduced and more interpretable feature set rather than the full 147-feature classification space. The selected variables included time_in_hospital, num_lab_procedures, num_procedures, num_medications, number_outpatient, number_emergency, number_inpatient, number_diagnoses, age, and the grouped diagnosis variables diag_1_group, diag_2_group, and diag_3_group. This produced a clustering dataset with 101,766 rows and 13 columns. The reduced feature set was chosen because it captured patient burden, healthcare utilisation, and diagnosis profile in a more interpretable way, while avoiding the sparsity and noise that would come from clustering on the full one-hot encoded modelling space. Before clustering, the numeric variables were scaled and the grouped diagnosis variables were indexed and encoded, resulting in a final clustering feature space of 52 variables.

6.2 Determining the number of clusters:

K Means was used to explore clustering structure in the reduced feature space because it provides a simple and widely used centroid based method for partitioning structured data (MacQueen, 1967). To determine an appropriate number of clusters, solutions from $k = 2$ to $k = 6$ were compared using both inertia and silhouette score. The silhouette scores were low overall, with the highest value at $k = 2$ (0.0658) and a very similar value at $k = 3$ (0.0631). This indicates that the clusters were not cleanly separated in a geometric sense, but this is not unusual in a healthcare dataset where patient profiles often overlap rather than forming sharply distinct groups.

Although $k = 2$ gave the highest silhouette score, $k = 3$ was selected as the final solution because it provided more informative subgroup structure while remaining very close in silhouette quality. In other words, the small drop in silhouette score was offset by a more useful and interpretable segmentation of the patient population. The low silhouette values were therefore acknowledged rather than ignored, and were interpreted as evidence of substantial overlap between patient subgroups rather than as proof that clustering was invalid. This made $k = 3$ the most appropriate balance between statistical fit and practical interpretability for the later local classification stage.

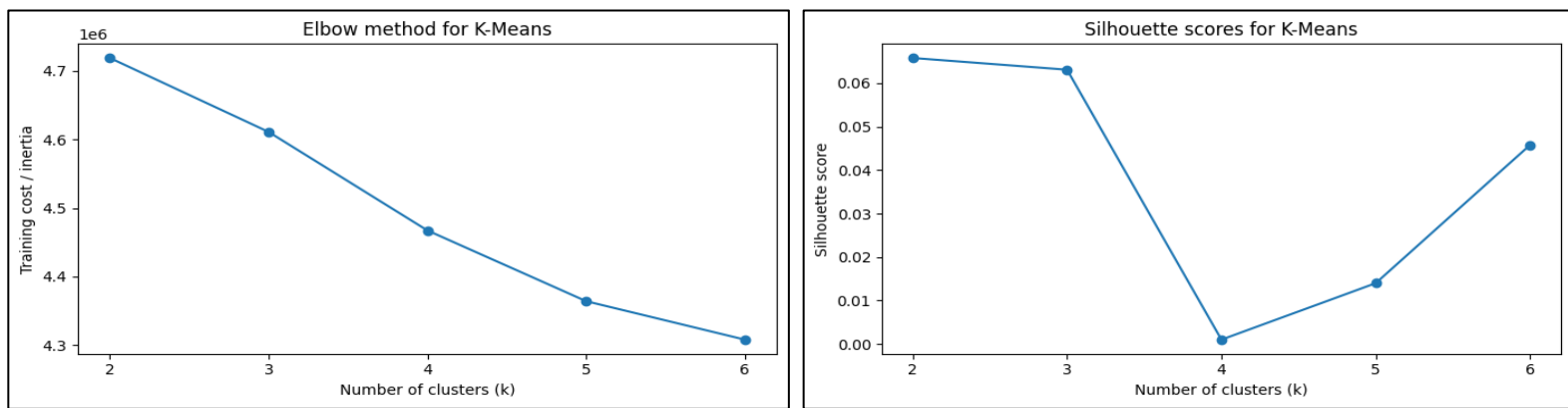


Figure 6. Elbow method (left) and silhouette scores (right) for candidate K Means solutions.

The final $k = 3$ solution produced clusters of size 3,930, 44,570, and 53,266. PCA visualisation showed partial separation, with the smallest cluster appearing more specific and the two larger clusters showing broader overlap. However, the first two principal components explained only about 9.7% of the total variance (5.43% for PC1 and 4.30% for PC2), so the PCA plot should be interpreted as a simplified visual aid rather than a complete representation of the full clustering structure. The readmission rates also differed modestly but consistently across clusters, which justified the subsequent evaluation of local cluster-based models.

6.3 Cluster structure and visualization:

The final clustering solution used $k = 3$, producing clusters of size 3,930, 44,570, and 53,266. Although the highest silhouette score was obtained at $k = 2$, $k = 3$ was retained because it provided a more interpretable subgroup structure for later profiling and comparison. To visualise the result, the clustering feature space was projected into two dimensions using principal component analysis. The PCA plot showed partial separation between the clusters, with one smaller cluster appearing more specific and the two larger clusters showing broader overlap. However, this visualisation should be interpreted with care because the first two principal components explained only about 9.7% of the total variance. The PCA plot is therefore useful as a simplified visual aid rather than as a complete representation of the full clustering structure. Overall, the visual pattern was consistent with the low silhouette scores, suggesting that the clusters captured meaningful subgroup tendencies but were not sharply separated in geometric terms.

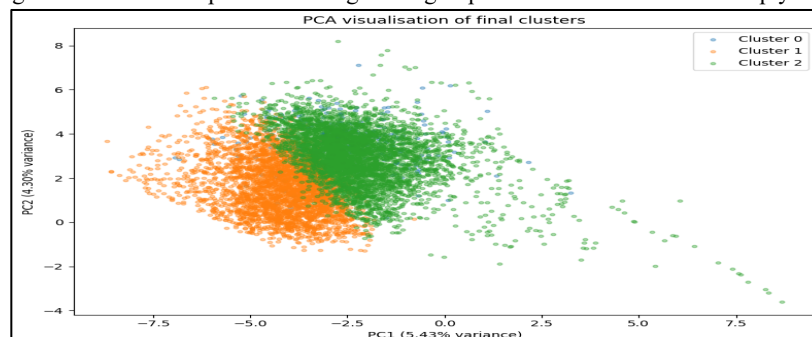


Figure 7. PCA visualisation of the final $k = 3$ clustering solution.

6.4 Cluster profiles:

The three clusters showed meaningful differences in both size and clinical profile. Cluster 0 contained 3,930 cases (3.86% of the dataset) and represented a smaller subgroup with a digestive-oriented diagnosis pattern and slightly higher readmission risk. Cluster 1 contained 44,570 cases (43.80%) and was characterised by higher treatment and procedure burden together with a circulatory-dominant diagnosis profile. Cluster 2 was the largest group, with 53,266 cases (52.34%), and represented a broader lower-intensity subgroup with generally lower treatment burden and a more mixed diagnosis pattern. These differences suggest that the clustering solution captured meaningful variation in patient burden and diagnosis context rather than producing purely arbitrary partitions.

Cluster	Size	Dataset share	Dominant profile	30-day readmission rate
0	3,930	3.86%	Smaller digestive-oriented subgroup with somewhat higher risk	11.96%
1	44,570	43.80%	Higher-treatment circulatory subgroup	11.45%
2	53,266	52.34%	Broader lower-intensity subgroup	10.85%

Table 6. Cluster profile summary

6.5 Readmission rates across clusters:

The readmission rates across the three clusters were different, although the gaps were moderate rather than dramatic. Cluster 0 had the highest 30-day readmission rate at 11.96%, followed by Cluster 1 at 11.45%, while Cluster 2 had the lowest rate at 10.85%. This shows that the clustering solution did capture some variation in early readmission risk, even though the differences were not large enough to suggest completely separate patient populations. These results were still important because they provided a practical justification for testing local cluster-based classifiers in the next stage. If readmission behaviour had been almost identical across all clusters, there would have been much less reason to expect value from local models.

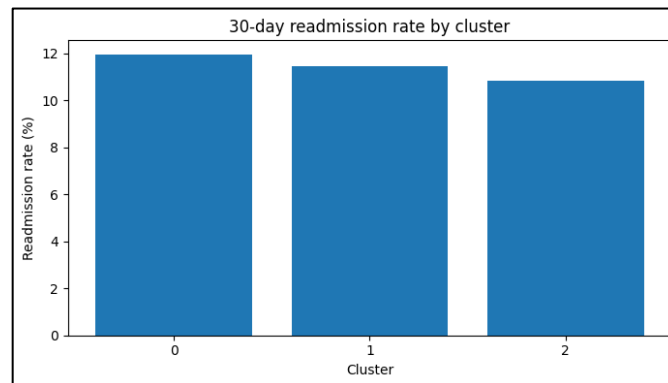


Figure 8. Thirty-day readmission rate by cluster.

6.6 Clinical interpretation of clusters:

The final clustering solution can be interpreted as three clinically meaningful subgroups. Cluster 0 represented a smaller digestive-oriented subgroup with slightly higher readmission risk. Cluster 1 represented a higher-treatment circulatory subgroup with greater procedure and medication burden. Cluster 2 represented a broader lower-intensity subgroup with the lowest readmission rate of the three. These labels do not imply perfectly distinct patient populations, but they do provide a useful summary of the main differences identified by the clustering stage and support the later evaluation of local cluster-based classifiers.

7. Cluster-based local classifiers

7.1 Local model setup:

Local classifiers were trained separately within each cluster in order to test whether subgroup-specific models could improve performance relative to a single global distributed model. Because the initial local models remained weak under the original class imbalance, each cluster's training data was balanced separately using undersampling logic consistent with the earlier modelling stage. This allowed a fairer assessment of whether cluster-specific training could improve minority-class detection within each subgroup.

7.2 Balanced local model results:

Balancing substantially improved the local models across all three clusters. The strongest local result came from Cluster 2, which achieved precision 0.1680, recall 0.5260, F1 score 0.2547, and ROC AUC 0.6453. Cluster 0 and Cluster 1 also improved after balancing, but both remained slightly weaker than Cluster 2. These results suggest that local modelling was most effective in Cluster 2, although the differences between the local models remained modest overall and did not translate into a clear advantage over the global weighted model.

7.3 Comparison of local and global models:

When compared directly with the global weighted Spark model, the balanced local models showed only limited predictive advantage. Cluster 2 slightly exceeded the global model on recall and ROC AUC, but not on F1 score. The global weighted model therefore remained the strongest distributed-stage model overall when judged on the balance between minority-class detection and overall stability. This indicates that clustering did not produce a clearly superior final classifier, even though it did reveal meaningful subgroup structure.

Model	Precision	Recall	F1	ROC-AUC
Global weighted Spark Logistic Regression	0.1695	0.5250	0.2563	0.6451
Local balanced model - Cluster 0	0.1486	0.4940	0.2284	0.6120
Local balanced model - Cluster 1	0.1574	0.4956	0.2389	0.6053
Local balanced model - Cluster 2	0.1680	0.5260	0.2547	0.6453

Table 7. Balanced local models compared with the global weighted distributed model

7.4 When cluster-based classification is beneficial:

Cluster-based classification is most likely to be beneficial when patient subgroups are more clearly separated than they were in this dataset, when individual clusters are large enough to support stable balanced training sets, and when the clusters differ more substantially in outcome risk than the modest differences observed here. In this project, the clusters differed in profile but not dramatically in readmission rate, which limited the practical advantage of local modelling. The main value of clustering was therefore interpretive rather than decisively predictive, since it helped reveal clinically meaningful subgroups even though the global weighted model remained slightly stronger overall.

8. Conclusion

This project developed a complete workflow for predicting whether a hospital encounter involving a diabetes patient would be followed by readmission within 30 days. The raw dataset was first audited and aggressively cleaned to produce a defensible modelling dataset, after which exploratory analysis showed that the strongest signals were associated with prior inpatient utilisation, discharge context, and treatment burden. The EDA also confirmed that the target variable was strongly imbalanced, making accuracy alone an unreliable basis for model evaluation.

In the initial modelling stage, all three baseline classifiers achieved high accuracy on the imbalanced data, but minority-class detection remained very weak. Random undersampling substantially improved recall and F1 score across all model families, and after tuning, the Random Forest achieved the strongest overall performance, with recall 0.6306, F1 score 0.2690, and ROC AUC 0.6590. This made the tuned Random Forest with random undersampling the strongest predictive model in the project.

The distributed stage extended the analysis by returning to the full raw dataset and reintroducing diagnosis information through grouped categories within a PySpark pipeline. Although the initial unweighted global Spark model performed poorly on minority-class detection, class weighting improved its performance substantially. The weighted Spark Logistic Regression became the strongest model within the distributed stage, but it still remained slightly weaker than the earlier tuned Random Forest overall. This showed that distributed processing added most value through richer feature retention, scalable preprocessing, and methodological extension rather than through a decisive improvement in final predictive performance. Clustering identified three interpretable patient subgroups with modest but meaningful differences in profile and readmission rate. Balanced local models improved considerably within these clusters, with Cluster 2 producing the strongest local result. However, the local models did not clearly outperform the global weighted distributed model overall. The main value of clustering was therefore interpretive rather than decisively predictive, since it revealed meaningful subgroup structure while offering only limited additional predictive benefit. Overall, the tuned Random Forest trained on the aggressively cleaned dataset with random undersampling provided the strongest final predictive performance, while the distributed and clustering stages added broader analytical value through richer data use, scalable modelling, and subgroup interpretation.

References

- Lemaître, G., Nogueira, F., & Aridas, C. K. (2017). Imbalanced-learn: A Python toolbox to tackle the curse of imbalanced datasets in machine learning. *Journal of Machine Learning Research*, 18(17), 1–5.
- MacQueen, J. (1967). Some methods for classification and analysis of multivariate observations. In *Proceedings of the fifth Berkeley symposium on mathematical statistics and probability* (Vol. 1, pp. 281–297). University of California Press.
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, E. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.
- Zaharia, M., Xin, R. S., Wendell, P., Das, T., Armbrust, M., Dave, A., Meng, X., Rosen, J., Venkataraman, S., Franklin, M. J., Ghodsi, A., Gonzalez, J., Shenker, S., & Stoica, I. (2016). Apache Spark: A unified engine for big data processing. *Communications of the ACM*, 59(11), 56–65.